Employee management system

A PROJECT REPORT ON

Employee Management System

A PARTIAL FULFILLMENT OF:

BCA SEM-6

SUBMITTED TO:

Geetanjali College of Computer Science and Commerce

AFFILIATED TO:

SAURASTRA UNIVERSITY

ACADEMIC YEAR:

2025-2026

UNDER THE GUIDANCE OF:

PROF. Jigisha Ganatra

PREPARED BY:

Davera Hardik

ACKNOWLEDGEMENT

While the report for my project work is ready, I feel it is my duty to offer my sincere thanks to all those who have helped and guided me in this regard.

Action is very necessary to convert our dreams into reality. My dream was to develop the project. To make a successful and useful project, one needs help, understanding, co-ordination from all those who are directly or indirectly involved in this.

Many people have contributed to make this project a reality. I would like to express my gratitude to our **Head of Department "Prof. Brijesh Shah"** for this guidance throughout the project.

While submitting the project report it is my duty to offer my sincere thanks to those who have assisted me in preparing this project report work.

Before I describe in brief about my project, I would like to add a few heartfelt words for those people who had given their time and knowledge to developing this project.

I am thankful to all my college faculties and friends for their kind co-operation.

Your Faith Fully,
Davera Hardik

PREFACE

In the 21st Century of Technology, Computers and related services are the most useful, automatic and highly speed of work. B.C.A is a course in which the practical aspect is as important as the theoretical aspect. In the present world, the importance of education is increasing. Project training in a student's life is like a live experience in the industry. If we have experience then & then we get top of the level.

The most important factor of a B.C.A Course is Project Work. The Project Work is Must for Every Student in a Different Government or Private Organization during the Study Period.

I am developing a project in the 6th Semester of B.C.A, I am a student of Geetanjali Group of Colleges affiliated with Saurashtra University, Rajkot.
I am creating this project dynamic and, in this project, we provide insert and delete operation on admin side and select at client side so the user can easily show the site.

I have gone through and studied necessary information in formation, which is collected and presented in this report. I am trying my best in this project. I have taken almost care that the information provided in the project is true and the project is in a smooth-running condition and error free.

PROJECT PROFILE

- Project Title :- Employee Management System
- Development Software :- Visual Studio 2022
- Front End :- Python
- Back End :- MySQL
- Academic Year :- 2025-2026
- Developed By :- Davera Hardik
- Submitted To :- Geetanjali Group of College
- Documentation Tool :- Microsoft Word
- Operating System :- Window 11

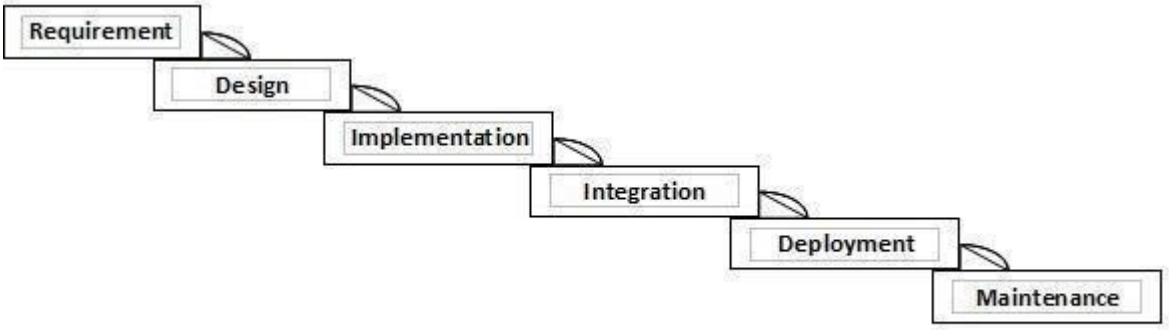
PROCESS MODEL

The Process Model used in our project "College Management System" is a waterfall model.

➢ The Waterfall Model:

The waterfall model is a sequential design process, used in software development processes, in which progress is seen as flowing steadily downwards (like a waterfall) through the phases of Conception, Initiation, Analysis, Design, Construction, Testing, Production/Implementation and Maintenance.

The waterfall development model originates in the manufacturing and construction industries: highly structured physical environments in which after-the-fact changes are prohibitively costly, if not impossible. Since no formal software development methodologies existed at the time, this hardware-oriented model was simply adapted for software development



Waterfall model

➢ Reason the waterfall model in the software development cycle:

- ➔ Since we have well known, clean and fixed requirements therefore it best suits our software development.
- ➔ Our product definitions are stable.
- ➔ Technology is clearly understood.
- ➔ The project is short.

➢ Advantages of the waterfall model:

- ➔ This model is simple and easy to understand and use. In this model phases are processed and completed one at a time and phases do not overlap.

- ➔ Waterfall model works well for smaller projects where sequence is very well understood

PROJECT INFORMATION

In the daily Employee Management System, we can use Python, MYSQL server Database and many packages. This is the project which can be used for Manage employee's role vice and handle their issues.

- **Admin Module:**
  1. **Register Employee:**
     In this section, Admin can only register new employees (add).
  2. **Employee:**
     In this section admin show employees details which were added by them and manage employees (edit, delete)
  3. **Leave:**
     In this section admin can approve leave and edit leave applications which employees registered by them and they can edit and delete their application.
  4. **Salary:**
     In this section admin can assign salaries to employees and also resolve corrections in salaries of employees.
  5. **Logout:**
     Through this section admin logout from the website

- **Employee:**
  1. **Profile:**
     In this section employees can view their details
  2. **Leave:**
     In this section can view their applied leaves status and edit their applied leaves if not have been accepted or rejected by superadmin and admins
  3. **Salary:**
     In this section employees can download their salary slip and also request for corrections if have mistaken in their salary.
  4. **Logout:**
     Through this section employees logout from system

SYSTEM REQUIREMENTS

- **Hardware Requirements:**

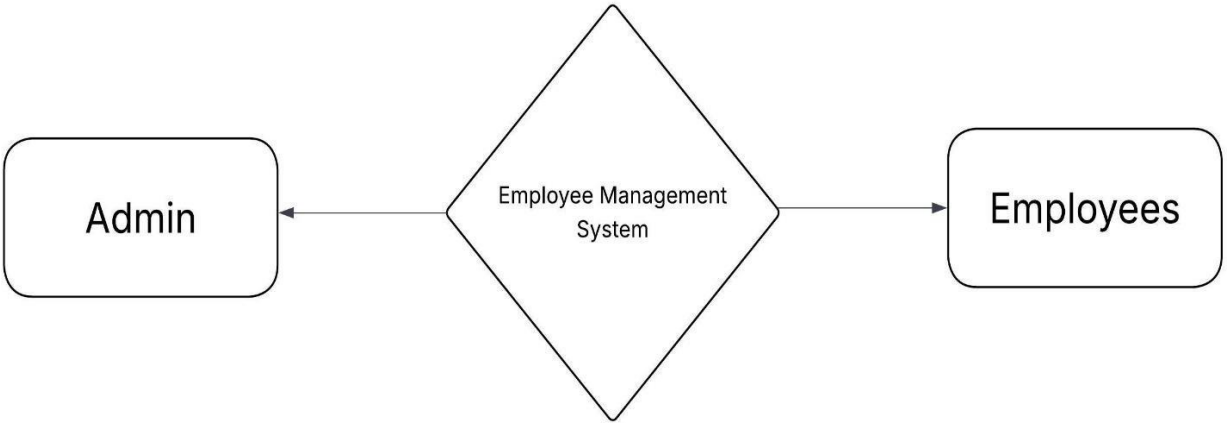
Processor	Intel(R) Core (TM) i5-835OU CPU @1.70Ghz 1.90Ghz
Ram	16.0GB (15.9GB) usable
Hard disk	64-bit operating system x64-based processor
Operating system	Windows-11

- **Software Requirements:**

Front End tool	Python
Back-end tool	MySQL
Editing tool	Visual Studio code
Browser	Internet Explorer or Mozilla Firefox or Google Chrome

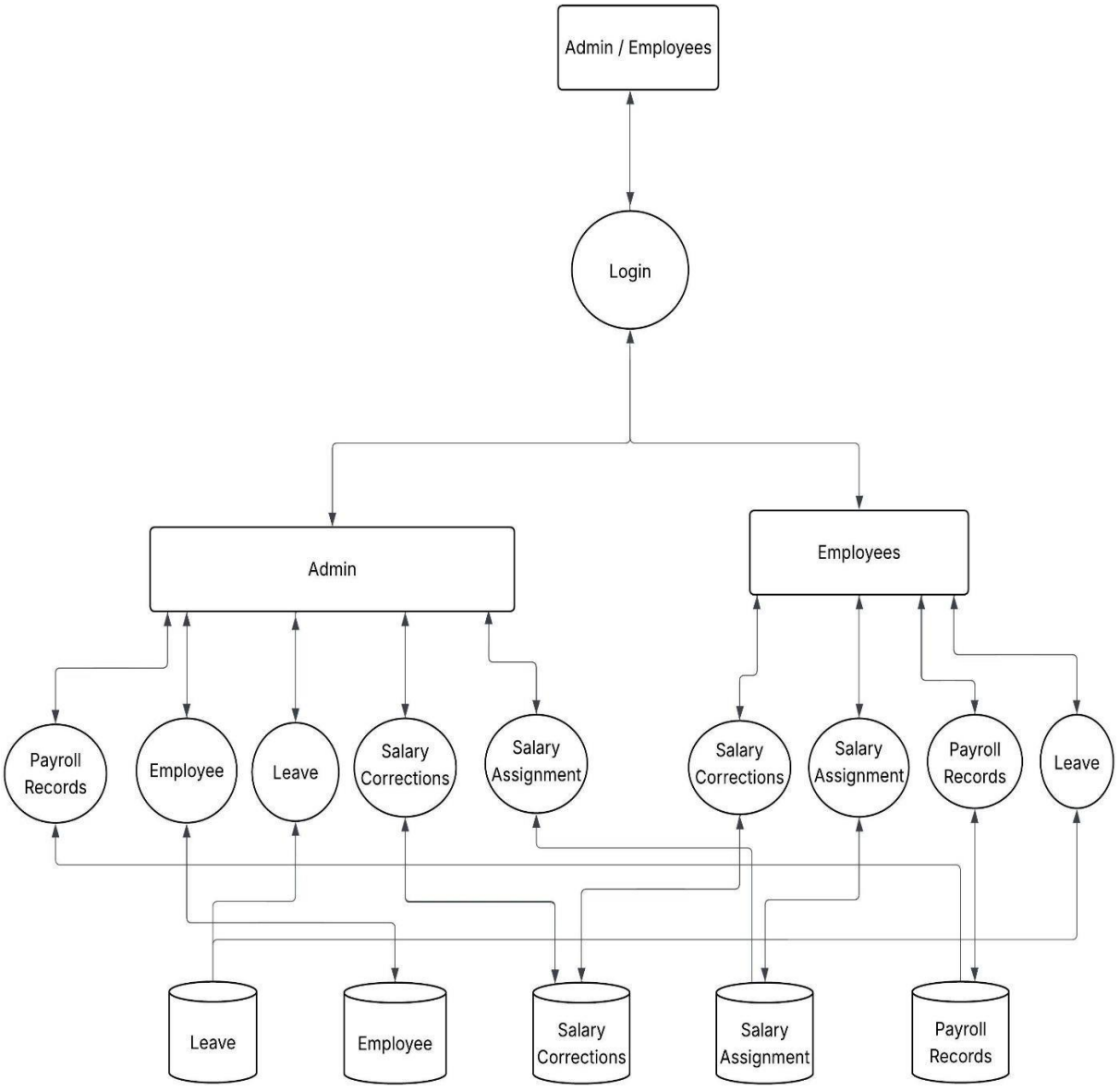
DATA FLOW DIAGRAM

- **CONTEXT LEVEL DFD: -**

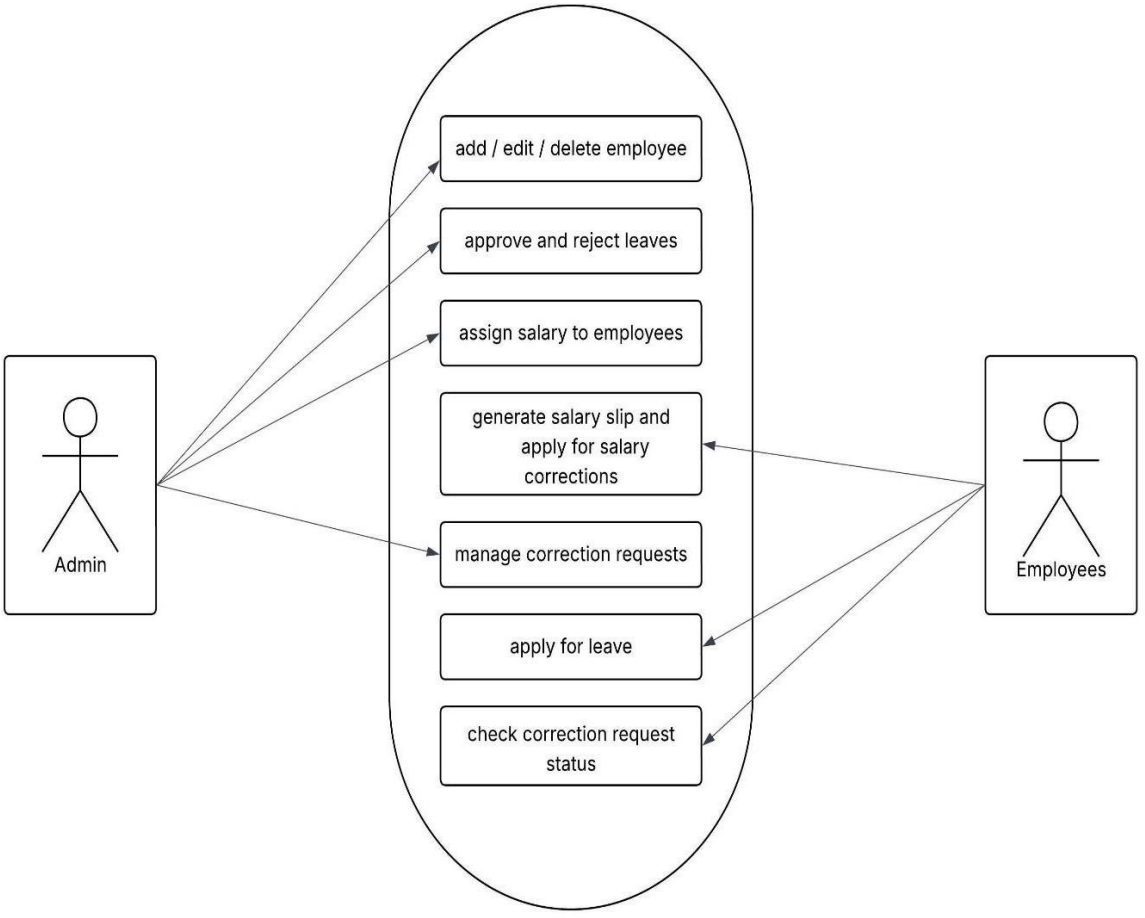


Employee management system

➢ ONE LEVEL DFD:



USE CASE DIAGRAM



DATA DICTIONARY

A data dictionary is a set of information describing the contents, format and structure of a database and the relationship between its elements, used to control access to and manipulation of the database.

- **DATABASE TABLES:**
  - employees
  - leaves
  - payroll_records
  - salary_assignments
  - salary_corrections

1. Employees:-

Field name	Data type	Size	Constraint key
Id	Varchar	10	PRIMARY KEY
Name	Varchar	50	Not NULL
Password	Varchar	50	Not NULL
Phone	Varchar	10	Not NULL
Email	Varchar	200	Not NULL
Salary	double	20	Not NULL
DOB	Date	-	Not NULL
Department	Varchar	10	Not NULL
Role	Varchar	10	Not Null

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2. Leaves:-

Field name	Data type	Size	Constraint key
Id	Varchar	10	PRIMARY KEY
emp_id	Varchar	50	Not NULL
emp_name	Varchar	50	Not NULL
Leave_type	Bigint	10	Not NULL
Start_date	date	-	Not NULL
End_date	Date	-	Not NULL
Reason	Text	-	Not NULL
Status	Varchar	32	Not NULL
Applied_date	Datetime	-	Not NULL
Duration_type	Varchar	32	Not Null

3. Salary_assignments:-

Fieldname	Data type	Size	Constraint key
Id	Varchar	64	PRIMARY KEY
Emp_id	Varchar	64	Not NULL
Assigned_salary	Double	-	Not NULL
Assigned_on	Datetime	-	Not NULL
Assigned_by	Varchar	64	Not NULL
Month	Varchar	7	Not NULL

4. Payroll_records:-

Fieldname	Data type	Size	Constraint key
Id	Varchar	64	PRIMARY KEY
Emp_id	Varchar	64	Not Null
Month	Varchar	7	Not Null
Base_salary	Double	-	Not Null
Overtime_hours	Double	-	Not NULL
Bonus	Double	-	Not NULL
Other_deductions	Double	-	Not NULL
Leave_deduction	Double	-	Not NULL
Net_salary	Double	-	Not NULL
Generated_on	Datetime	-	Not NULL
Slip_path	Varchar	255	Not NULL

5. Salary_corrections:-

Fieldname	Data type	Size	Constraint key
Id	Varchar	64	PRIMARY KEY
Emp_id	Varchar	64	Not NULL
Month	Varchar	7	Not NULL
Description	Text	-	Not NULL
Submitted_on	Datetime	-	Not NULL
Status	Varchar	32	Not NULL
Assignment_id	Varchar	64	Not NULL
Payroll_id	Varchar	64	Not NULL

TEST CASES

The testing phase is an important part of the software developed. It is the process of finding errors and missing operations and a complete verification to determine whether the objectives are met and the user requirements are satisfied.

➢ Login Screen:

Test Case ID	Module	Test Scenario	Test Steps	Input Data	Expected Result	Actual Result	Status
TC001	Login	Valid Login	Enter valid admin id& password	Admin id: EMP001 Password: admin@123	User redirectedto Home page	As expected,	Pass
TC002	Login	Invalid login	Enter valid id & wrong password	Admin id: EMP001, Password: xyz	Error message shown	As expected,	Pass
TC003	Login	Valid Login	Enter valid id & password	User id: EMP002 Password: user@123	User redirect to Login page	As expected,	Pass
TC004	Login	Invalid Login	Enter valid id & password	User id: EMP002 Password:xyz	Error message shown	As expected,	Pass

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➢ Employee Register Screen for Admins:

Test Case ID	Module	Test Scenario	Test Steps	Input Data	Expected Result	Status
TC001	Register	Successful Registration	Fill all fields and click "Register"	Name, Phone, Email, Password ,Date of Birth, Departme nt, role, salary	Account created & confirmation message shown	Pass
TC002	Register	Missing Field Validations	Leave "Email" and 'phone'', ''Password'' empty and click "Register"	Name, Date of Birth, Departme nt, role, salary	Error message: "Fields are required"	Pass
TC003	Register	Duplicate Email and contact no	Try to register with an already registered email and contact number	Email: test@test .com phone: 12345678 90	Error message: "Email and phone already exists"	Pass
TC004	Register	Invalid Joining Date and Birth of Date	Date of Birth is same, If Birth of date is invalid means age is under 18	Date of Birth: 28-9-16	Error Message : ''Enter Valid dates"	Pass

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➢ Employees Listing for Admins:

Test Case ID	Module	Test Scenario	Test Steps	Expected Result	Status
TC001	Employee list	Employees List	Admins: employees listed here which are registered by them.	List of employees and admins shown basis on role	Pass

➢ Show Leave Applications Screen:

Test Case ID	Module	Test Scenario	Test Steps	Expected Result	Status
TC001	Leave (Admin)	Leave Applications List	Admins: Which employees leave applications listed here which are registered by them.	Leave Applications list which applied by employees	Pass

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➢ Employees Edit Screen for Admins:

Test Case ID	Module	Test Scenario	Test Steps	Input Data	Expected Result	Status
TC001	Edit	Successful Edit	Fill all fields and click "Edit"	Name, phone, Email, Password, Date of Birth, Departme nt, role, salary	Account Edit & confirmation message shown	Pass
TC002	Edit	Missing Field Validations	Leave "Email" and 'phone'', ''Password'' empty and click "Edit"	Name, Date of Birth, Departme nt, role, salary	Error message: "Fields are required"	Pass
TC003	Edit	Duplicate Email and contact no	Try to Edit with an already registered email and phone	Email: test@test.com phone: 1234567890	Error message: "Email and phone already exists"	Pass
TC004	Edit	Invalid Birth of Date	If Birth of date is invalid means age is under 18	Date of Birth: 28-9-16	Error Message: ''Enter Valid dates"	Pass

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➢ Leave Screen:

Test Case ID	Module	Test Scenario	Test Steps	Input Data	Expected Result	Status
TC001	Leave	Leave Request	Fill Leave Form and click "Apply Leave"	Leave type, leave reason, from date and to date	Leave request submitted successfully and send message	Pass
TC002	Leave	Missing leave type, From and to date	Submit without leave type and from and to date	Leave reason	Error Message: Fill required fields please	Pass
TC003	Leave	Invalid Date range	From date greater than to date	From: 29-9-25 To: 28-9-25	Error Message: Enter Valid dates	Pass
TC004	Leave	Change status	Leave must be approved or rejected	Leave status will be accepted or rejected	Success Message: Leave Status change successfully	Pass

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TC005	Leave	Edit Leave	Leave will not be accepted or rejected	Leave type, Leave Reason, From date and to date	Success Message: Leave Updated successfully	Pass
TC006	Leave	Edit & Delete Leave	Leave status will be pending	Edit: Leave Type, Leave reason, from date, To date	Success Message: Edit -> Leave Application Updated successfully, Delete -> Leave application deleted successfully	Pass

REFERENCE AND BIBLIOGRAPHY

➢ Websites: -

-http://www.apachefreinds.org

-http://www.mysql.com

➢ Search engines: -

-Google search engines
-Mozilla firefox
-Google chrome

➢ Reference engines: -

-www.google.com

-www.w3schools.com